

Regime-Aware Machine Learning for Technical Pattern Trading

Event-Based Classification and Profitability Evaluation on SPY

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Problem Motivation

Why financial ML needs event-based approaches

- **Financial markets are extremely noisy** — most bars contain no actionable signal
- **Naive price prediction** models learn noise, price level, or time trends
- **Event-based learning** focuses on *technically meaningful moments*

Approach	Prediction Target	Key Problem
Naive regression	Next bar's close price	Learns noise & trends
Bar-level classification	Up/down per bar	No trading context
Event-based (this work)	Trade outcome at signal bars	Focused, actionable

Core question: *When price is at a technically meaningful structure, can we predict the outcome of a trade-sized move?*

Background

Foundations of the approach

- **Technical analysis patterns:** support/resistance, channels, triangles, multiple tops/bottoms
- **Triple-barrier labeling** (Lopez de Prado, 2018): labels based on trade outcomes, not arbitrary direction
- **Random Forest / Bagging:** ensemble tree models — robust, interpretable, handle mixed features
- **Walk-forward validation:** simulates realistic deployment with expanding training windows
- **Profitability evaluation:** backtesting with TP/SL/time exits — accuracy alone is insufficient

Key insight from financial ML literature: the best classifier is not necessarily the best trading model. Profitability depends on the *magnitude* of correct predictions, not just their frequency.

System Architecture

End-to-end pipeline from raw data to profitability evaluation



Stage	Module	Output
Data	load_data.py	4,023 SPY daily bars (2010-2025)
Detection	scanner.py + 4 detectors	132 event signals
Features	indicators.py + build_features.py	48 features per event
Labeling	label_events.py	long / short / no_trade labels
Training	train.py + optimize.py	RF, Bagging, Baseline models
Evaluation	simulator.py + train.py	Classification + profitability metrics

Pattern Detection

Four complementary detectors identify trading opportunities

Detector	Method	Signals	Key Parameters
Support / Resistance	Rolling max/min + stability filter	42 events	window=50, $0.3 \times \text{ATR}$, cooldown=10
Triangles	Swing pivots + linear regression	17 events	window=25, $ r \geq 0.85$, convergence $\geq 5\%$
Channels	Chunk extremes + polyfit	12 events	lookback=40 $\pm$ 15, $\geq 2+3$ touches
Multiple Tops/Bottoms	Rolling extremes + trend confirm	63 events	window=20, 5-bar slope confirm
TOTAL		132 events	3.3% event density

- All detectors use **cooldown filters** (10 bars) to prevent signal clustering
- Signal localized at the **event bar** — the bar where the pattern condition is met
- Per supervisor feedback: triangles and channels excluded from training labels

Feature Engineering

48 features with strict leakage prevention

Feature Group	Count	Examples
Volatility & Returns	9	atr_ratio, rvol_20, ret_1/5/10/20, mom_5/10/20
Moving Average Distances	8	sma_10_dist ... sma_200_dist, MA spreads
Momentum Indicators	4	rsi_14, macd_norm, macd_signal_norm, macd_hist_norm
Bollinger Bands	2	bb_width, bb_pctb
Binary Technical Filters	8	bb_touch_upper/lower, SMA cross, RSI extremes
Volume	2	volume_ratio, volume_std
Pattern Geometry	11	slopes, touches, containment, width, R ²
Event Type Dummies	4–6	One-hot encoded pattern type
TOTAL	48–50	

Removed (trend-leaking): raw ATR, absolute SMA values, cumulative OBV, raw MACD. All features use relative, normalized, or bounded values only.

Triple-Barrier Labeling & Optimization

TP/SL/max_holding treated as tunable hyperparameters

- **Triple-barrier method:** for each event, walk forward and check barriers

Barrier	Definition	Label if hit first
Upper (TP)	$\text{entry_price} + \text{pt_mult} \times \text{ATR}$	long
Lower (SL)	$\text{entry_price} - \text{sl_mult} \times \text{ATR}$	short
Time	max_holding bars elapsed	no_trade

Innovation: pt_mult, sl_mult, max_holding are *not fixed* — they are optimized via grid search (100 configurations).

Search Space	Range	Best for F1	Best for Profit
pt_mult	1.0 – 3.0	2.0	2.5
sl_mult	1.0 – 3.0	1.5	3.0
max_holding	5 – 20 bars	10	20
Score		F1 = 0.569	Return = 25.9%

Machine Learning & Validation

Three models, three validation strategies

- **Models:** Random Forest (200 trees), Bagging (200 trees), Stratified Baseline
- **Class weighting:** balanced (adjusts for label imbalance)

Validation Method	Description	Purpose
Chronological split	60% train / 20% val / 20% test	Primary evaluation
Walk-forward CV	5 expanding folds, temporal order	Realistic deployment simulation
5-fold event CV	Rotate test fold, 80/20 train/val	Reduce sampling noise (diagnostic)
Tree diagnostics	Per-tree accuracy, ensemble gain	Detect ensemble dependency

- Walk-forward CV respects temporal order — no future data in training
- Individual tree analysis shows whether ensemble relies on few strong trees or distributes strength

Trading Simulation & Profitability

Why accuracy alone is insufficient

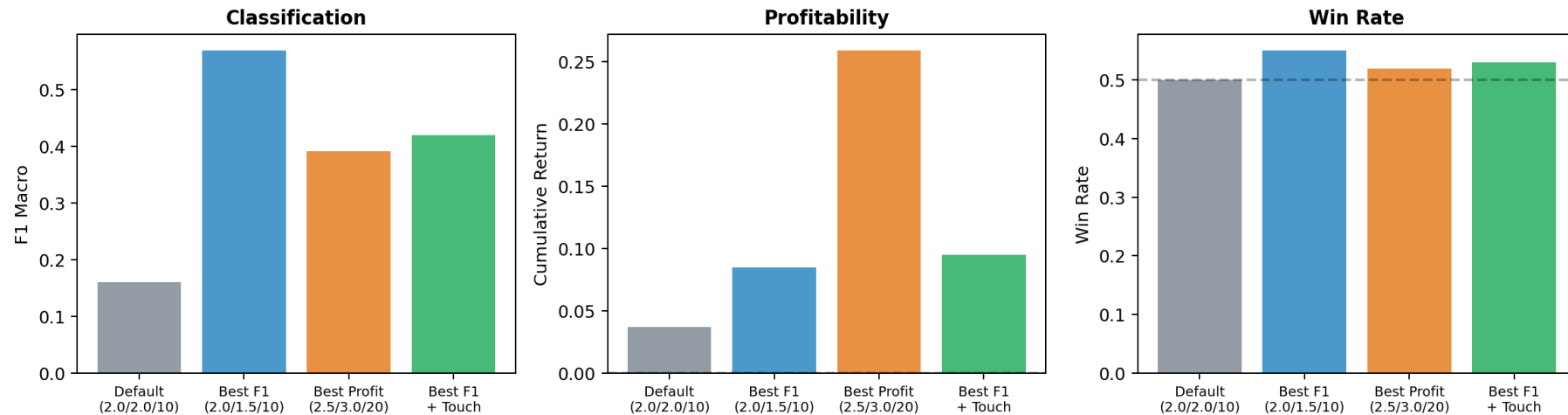
- **Simulated trading:** enter at signal-bar close, exit at TP/SL/time barrier
- A model with **lower accuracy** can be **more profitable** if correct predictions capture larger moves

Metric	Definition	Interpretation
Cumulative return	$\Sigma(\text{net returns})$	Total P&L
Win rate	% trades with positive return	Hit rate
Profit factor	Gross profit / gross loss	Reward-to-risk ratio
Sharpe ratio	Mean return / std(returns)	Risk-adjusted performance
Max drawdown	Largest peak-to-trough decline	Worst-case scenario

Assumptions: no transaction costs, no slippage, equal position sizing, no compounding. Entry at signal-bar Close (consistent with labeling pipeline).

Experimental Results

104 detector events + 38 touch events = 142 total



Configuration	pt / sl / mh	F1 Macro	Cum. Return	Win Rate	Trades
Default (baseline)	2.0 / 2.0 / 10	0.160	3.7%	50%	18
Best for F1	2.0 / 1.5 / 10	0.569	8.5%	55%	18
Best for Profit	2.5 / 3.0 / 20	0.392	25.9%	52%	15
Best F1 + Touch	2.0 / 1.5 / 10	0.420	9.5%	53%	24

Key finding: optimal parameters for classification (F1) differ from those for profitability (return). This confirms that trading evaluation requires profitability metrics beyond accuracy.

Limitations

Honest assessment of constraints

- **Small dataset:** ~140 events provides limited statistical power; high result variance
- **Single asset:** only SPY tested — generalization to other markets is unknown
- **No transaction costs:** spread, slippage, and commissions are not modeled
- **Simplified entry:** signal-bar Close, not next-bar Open (slight optimism)
- **Optimization overfitting risk:** 100 configurations on ~100 events
- **Touch-event noise:** additional events may not carry the same signal quality as strict detectors
- **No regime modeling:** HMM regime detection planned but not yet integrated

All results should be interpreted as *preliminary research findings*, not production-ready trading signals.

Conclusion & Future Work

What was achieved:

- End-to-end pipeline: data → detection → features → labeling → ML → validation → backtest
- 48 leakage-free features from 4 pattern detectors on 4,023 SPY bars
- Triple-barrier parameters converted from fixed constants to tunable hyperparameters
- Profitability evaluation integrated into training and cross-validation
- 3.6× F1 improvement over baseline; 25.9% cumulative return (best profit config)

Main scientific lesson:

- Classification accuracy and trading profitability optimize at *different* parameter settings

Future work:

- Multi-asset testing, transaction cost modeling, purged/embargo CV
- Regime-aware dynamic TP/SL, HMM integration, more advanced models